Havirbhavi Pothugunta

Portland, OR | +1 (503) 406-7543 | havirbha@pdx.edu | linkedin.com/in/havirbhavi-pothugunta | github.com/Havirbhavi

Summary

Machine Learning Engineer with experience building end-to-end ML pipelines, LLM/RAG features, backend APIs, and cloud-deployed systems. Skilled in Python, FastAPI, PostgreSQL, AWS/GCP, and deep learning frameworks, with hands-on experience in data ingestion, model evaluation, dashboard delivery, and production-ready validation workflows.

Education

Portland State University — Portland, OR
Master of Science in Computer Science; GPA: 3.89 / 4.00 — *Sep 2024 – Mar 2026*

Amrita Vishwa Vidyapeetham — Chennai, India
Bachelor of Technology in Computer Science Engineering; GPA: 8.08 / 10 — *Oct 2020 – May 2024*

Experience

Growly — USA
Co-Founder, Full-Stack Engineer — *2025*

- Developed a two-step LLM pipeline to monitor Gmail job-application emails, classify subjects, and extract structured fields such as job title, company, status, and date into PostgreSQL.
- Implemented high-concurrency ingestion with batching, validation, deduplication, idempotent writes, and retry handling to reduce duplicate and missing records.
- Built FastAPI services and a React dashboard; deployed backend infrastructure on AWS RDS and supported onboarding for 200+ beta users with 100 active users.

Amrita School of Computing — Chennai, India
Research Assistant, Deep Learning — *Aug 2023 – Nov 2024*

- Designed a 5-class coffee leaf disease classification system using ResNet50 and MobileNet with transfer learning for image-based plant disease detection.
- Applied augmentation techniques including rotation, flipping, and zooming to improve robustness and achieved approximately 99.9% validation accuracy.
- Presented research findings at an international conference and contributed to publications across IEEE and Springer venues.

Projects

AdventureLLM: LLM Benchmarking for Sequential Decision-Making | *Python, OpenAI API, CSV Telemetry* — 2025

- Implemented a modular Zork benchmarking pipeline to evaluate LLM agents across repeated episodes and runs in a text-based decision-making environment.
- Logged per-turn telemetry including score, moves, inventory, token usage, and model responses to CSV for repeatable analysis and comparison.
- Benchmarked 5 LLMs and analyzed score distributions, identifying common failure modes such as invalid commands, repetitive loops, and context-limit issues.

GeoTransit Analytics: TriMet Cloud Data Engineering Pipeline | *GCP Pub/Sub, GCP VM, PostgreSQL* — 2025

- Designed a cloud data pipeline to ingest TriMet GPS telemetry from Jan. 1–8, 2023, processing 35–40M rows and approximately 40GB of transit data.
- Built Pub/Sub-based producer and consumer services on separate GCP VMs with cron scheduling to support automated message ingestion.
- Enforced one-message-one-row processing with 20 validation rules and generated reporting-ready metrics for transit performance dashboards.

Skills

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, Data Structures
ML / GenAI: Pandas, Visualization, PyTorch, TensorFlow/Keras, RAG, Prompt Engineering, LLMs, XGBoost, NLP
Backend & Databases: FastAPI, REST APIs, Spring Boot, PostgreSQL, Node.js
Cloud & Tools: AWS SQS, AWS S3, AWS RDS, GCP Pub/Sub, Terraform, DVC, Docker, Kubernetes, CI/CD, Git

Publications

IEEE ICESIC 2024 | IEEE INOCON 2024 | Springer ICDSNE 2024 | Springer ICCIDS 2024